Examiner only

10. *(a)* Butylamine can be made from propan-1-ol in a three-stage process.

Give the reagent used and the structural formula of the product for each stage. [3]

Stage	Reagent used	Structural formula of product
1		
2		
3		$CH_3CH_2CH_2CH_2NH_2$

(b) Below 10 °C, phenylamine reacts with nitric(III) acid, HNO_2, to give a benzenediazonium compound, which can react with phenol to give an azo dye.

(i) State how nitric(III) acid is prepared for use in this reaction. [1]

....

....

(ii) Give the equation for the reaction of benzenediazonium chloride with phenol to produce an azo dye. [1]

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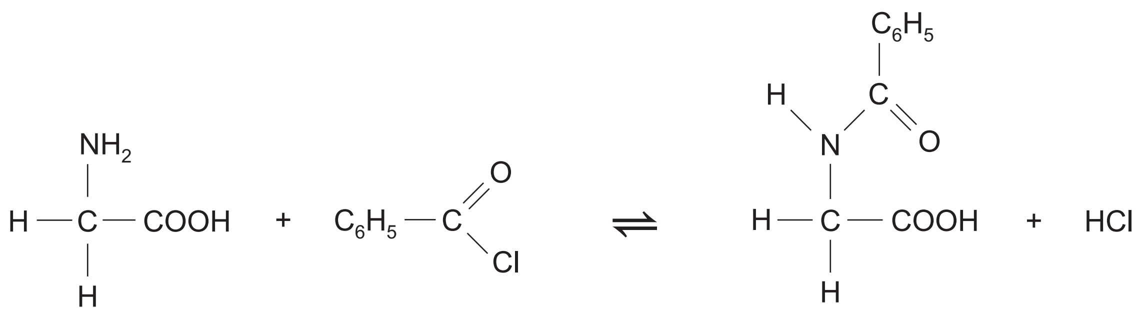
(c) Hippuric acid, so called because it is found in the urine of horses, can be made by reacting aminoethanoic acid and benzoyl chloride.



benzoyl chloride hippuric acid

In an experiment the percentage yield was 90%.

Benzoyl chloride is only slowly hydrolysed by alkalis. A teacher suggested the following idea for increasing the percentage yield.

'The experiment could be carried out in the presence of aqueous sodium hydroxide.'

Comment on the validity of this suggestion and any effect it may have on the yield. Explain your answer. [1]

....

....

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(d) A straight chain α-amino acid of general formula $R—CH(NH_2)COOH$ reacts with nitric(III) acid to give nitrogen gas.

$$R—CH(NH_2)COOH \xrightarrow{HNO_2} R—CH(OH)COOH + N_2 + H_2O$$

In an experiment 7.87 g of the amino acid was dissolved in water and the solution made up to 500 cm^3. A 25.0 cm^3 sample of this solution was reacted with an excess of nitric(III) acid and gave 73.5 cm^3 of nitrogen measured at 298 K and 1 atm pressure.

Calculate the molar mass of the amino acid and hence its structural formula. [4]

Molar mass = ...

Structural formula

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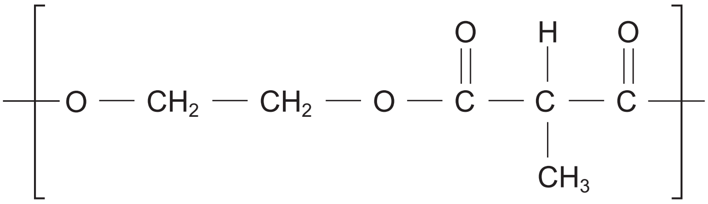
(e) Polyesters can be made from a dicarboxylic acid and a diol, such as ethane-1,2-diol.



polyester **E**

(i) State the name of the dicarboxylic acid that reacts with ethane-1,2-diol to give polyester **E**. [1]

....

(ii) Complete the table to describe the high resolution ^{1}H NMR spectrum of polyester **E**. [3]

Protons	Splitting pattern	Relative peak area
$—CH_2—CH_2—$		
$—CH—$		
$—CH_3$		

14

END OF PAPER

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